

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{Semaphore} \times \text{Semaphore} \times \text{int} \times \text{int} \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{int} \times \text{int} \times \text{int} \Rightarrow \text{Process} \\ \text{sem} : \text{int} \times \text{int} \times \text{int} \Rightarrow \text{Semaphore} \\ \text{bbr2} : \text{Fcall} \\ \text{bbr3} : \text{Fcall} \\ \text{bbr4} : \text{Fcall} \\ \text{bbr5} : \text{Fcall} \\ \text{bbr6} : \text{Fcall} \\ \text{cmr2} : \text{Fcall} \\ \text{cmr3} : \text{Fcall} \\ \text{cmr4} : \text{Fcall} \\ \text{cmr5} : \text{Fcall} \\ \text{cmr6} : \text{Fcall} \\ \text{cmr7} : \text{Fcall} \\ \text{cmr11} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1 :	cnfg(p(bbr2, nb, 0, ns), pc, semb, sem(sc, dc, tc), sems, fs, 0) → cnfg(p(bbr3, nb, 0, ns), pc, semb, sem(sc1, dc, tc), sems, fs, 0)	[$\neg sc = 0 \wedge sc1 = sc - 1$]
2 :	cnfg(p(bbr2, nb, 0, ns), pc, semb, sem(sc, dc, tc), sems, fs, 0) → cnfg(p(bbr2, nb, tc, ns), pc, semb, sem(sc, dc, tc2), sems, fs, 0)	[$sc = 0 \wedge tc2 = tc + 2$]
3 :	cnfg(p(bbr2, nb, nc, ns), pc, semb, sem(sc, dc, tc), sems, fs, 1) → cnfg(p(bbr3, nb, 0, ns), pc, semb, sem(sc, dc, tc), sems, fs, 0)	[$\neg nc = 0 \wedge nc = dc$]
4 :	cnfg(p(bbr3, nb, nc, 0), pc, semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(p(bbr4, nb, nc, 0), pc, semb, semc, sem(ss1, ds, ts), fs, 0)	[$\neg ss = 0 \wedge ss1 = ss - 1$]
5 :	cnfg(p(bbr3, nb, nc, 0), pc, semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(p(bbr3, nb, nc, ts), pc, semb, semc, sem(ss, ds, ts2), fs, 0)	[$ss = 0 \wedge ts2 = ts + 2$]
6 :	cnfg(p(bbr3, nb, nc, ns), pc, semb, semc, sem(ss, ds, ts), fs, 1) → cnfg(p(bbr4, nb, nc, 0), pc, semb, semc, sem(ss, ds, ts), fs, 0)	[$\neg ns = 0 \wedge ns = ds$]
7 :	cnfg(p(bbr4, nb, nc, ns), pc, semb, semc, sems, fs, 0) → cnfg(p(bbr5, nb, nc, ns), pc, semb, semc, sems, fs1, 0)	[$fs1 = (fs + 1)$]
8 :	cnfg(p(bbr5, 0, nc, ns), pc, sem(sb, db, tb), semc, sems, fs, 0) → cnfg(p(bbr6, 0, nc, ns), pc, sem(sb, db2, tb), semc, sems, fs, 1)	[$\neg tb = db + 2 \wedge db2 = db + 2$]
9 :	cnfg(p(bbr5, 0, nc, ns), pc, sem(sb, db, tb), semc, sems, fs, 0) → cnfg(p(bbr6, 0, nc, ns), pc, sem(sb1, db, tb), semc, sems, fs, 0)	[$tb = db + 2 \wedge sb1 = sb + 1$]
10 :	cnfg(p(bbr6, nb, nc, 0), pc, semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(p(bbr2, nb, nc, 0), pc, semb, semc, sem(ss, ds2, ts), fs, 1)	[$\neg ts = ds + 2 \wedge ds2 = ds + 2$]
11 :	cnfg(p(bbr6, nb, nc, 0), pc, semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(p(bbr2, nb, nc, 0), pc, semb, semc, sem(ss1, ds, ts), fs, 0)	[$ts = ds + 2 \wedge ss1 = ss + 1$]
12 :	cnfg(pb, p(cmr2, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr3, nb, nc, 0), semb, semc, sem(ss1, ds, ts), fs, 0)	[$\neg ss = 0 \wedge ss1 = ss - 1$]
13 :	cnfg(pb, p(cmr2, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr2, nb, nc, ts), semb, semc, sem(ss, ds, ts2), fs, 0)	[$ss = 0 \wedge ts2 = ts + 2$]
14 :	cnfg(pb, p(cmr2, nb, nc, ns), semb, semc, sem(ss, ds, ts), fs, 1) → cnfg(pb, p(cmr3, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0)	[$\neg ns = 0 \wedge ns = ds$]
15 :	cnfg(pb, p(cmr3, nb, nc, ns), semb, semc, sems, fs, 0) → cnfg(pb, p(cmr4, nb, nc, ns), semb, semc, sems, fs, 0)	[$fs > 0$]
16 :	cnfg(pb, p(cmr3, nb, nc, ns), semb, semc, sems, fs, 0) → cnfg(pb, p(cmr11, nb, nc, ns), semb, semc, sems, fs, 0)	[$fs \leq 0$]
17 :	cnfg(pb, p(cmr4, nb, nc, ns), semb, semc, sems, fs, 0) → cnfg(pb, p(cmr5, nb, nc, ns), semb, semc, sems, fs1, 0)	[$fs1 = (fs - 1)$]
18 :	cnfg(pb, p(cmr5, nb, 0, ns), semb, sem(sc, dc, tc), sems, fs, 0) → cnfg(pb, p(cmr6, nb, 0, ns), semb, sem(sc, dc2, tc), sems, fs, 1)	[$\neg tc = dc + 2 \wedge dc2 = dc + 2$]
19 :	cnfg(pb, p(cmr5, nb, 0, ns), semb, sem(sc, dc, tc), sems, fs, 0) → cnfg(pb, p(cmr6, nb, 0, ns), semb, sem(sc1, dc, tc), sems, fs, 0)	[$tc = dc + 2 \wedge sc1 = sc + 1$]
20 :	cnfg(pb, p(cmr6, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr7, nb, nc, 0), semb, semc, sem(ss, ds2, ts), fs, 1)	[$\neg ts = ds + 2 \wedge ds2 = ds + 2$]
21 :	cnfg(pb, p(cmr6, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr7, nb, nc, 0), semb, semc, sem(ss1, ds, ts), fs, 0)	[$ts = ds + 2 \wedge ss1 = ss + 1$]
22 :	cnfg(pb, p(cmr7, 0, nc, ns), sem(sb, db, tb), semc, sems, fs, 0) → cnfg(pb, p(cmr2, 0, nc, ns), sem(sb1, db, tb), semc, sems, fs, 0)	[$\neg sb = 0 \wedge sb1 = sb - 1$]
23 :	cnfg(pb, p(cmr7, 0, nc, ns), sem(sb, db, tb), semc, sems, fs, 0) → cnfg(pb, p(cmr7, tb, nc, ns), sem(sb, db, tb2), semc, sems, fs, 0)	[$sb = 0 \wedge tb2 = tb + 2$]
24 :	cnfg(pb, p(cmr7, nb, nc, ns), sem(sb, db, tb), semc, sems, fs, 1) → cnfg(pb, p(cmr2, 0, nc, ns), sem(sb, db, tb), semc, sems, fs, 0)	[$\neg nb = 0 \wedge nb = db$]
25 :	cnfg(pb, p(cmr11, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr2, nb, nc, 0), semb, semc, sem(ss, ds2, ts), fs, 1)	[$\neg ts = ds + 2 \wedge ds2 = ds + 2$]
26 :	cnfg(pb, p(cmr11, nb, nc, 0), semb, semc, sem(ss, ds, ts), fs, 0) → cnfg(pb, p(cmr2, nb, nc, 0), semb, semc, sem(ss1, ds, ts), fs, 0)	[$ts = ds + 2 \wedge ss1 = ss + 1$]
27 :	cnfg(pb, pc, semb, semc, sems, fs, a) → success	
28 :	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
29 :	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
30 :	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
31 :	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
32 :	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
33 :	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
34 :	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
35 :	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
36 :	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
37 :	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
38 :	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
39 :	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
40 :	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
41 :	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
42 :	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	

$$\{43: \text{cnfg}(p(bbr2, 0, 0, 0), p(cnr2, 0, 0, 0), \text{sen}(0, db, tb), \text{sen}(0, dc, tc), \text{sen}(1, ds, ts), 1, 0) \Rightarrow \text{succss} [tb = db + 2 \wedge (db > 0) \wedge tc = dc + 2 \wedge (dc > 0) \wedge ts = ds + 2 \wedge (ds > 0)]\}$$

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{Semaphore} \times \text{Semaphore} \times \text{bitvec}_1 \times \text{bitvec}_1 \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{bitvec}_3 \times \text{bitvec}_3 \times \text{bitvec}_3 \Rightarrow \text{Process} \\ \text{sem} : \text{bitvec}_1 \times \text{bitvec}_3 \times \text{bitvec}_3 \Rightarrow \text{Semaphore} \\ \text{bbr2} : \text{Fcall} \\ \text{bbr3} : \text{Fcall} \\ \text{bbr4} : \text{Fcall} \\ \text{bbr5} : \text{Fcall} \\ \text{bbr6} : \text{Fcall} \\ \text{cmr2} : \text{Fcall} \\ \text{cmr3} : \text{Fcall} \\ \text{cmr4} : \text{Fcall} \\ \text{cmr5} : \text{Fcall} \\ \text{cmr6} : \text{Fcall} \\ \text{cmr7} : \text{Fcall} \\ \text{cmr11} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1:	cnfg(p(bbr2, nb, #b000, ns), pc, semb, sem(sc, dc, tc), sems, fs, #b0) → cnfg(p(bbr3, nb, #b000, ns), pc, semb, sem(sc1, dc, tc), sems, fs, #b0)	[¬sc = #b0 ∧ sc1 = scbvsub#b1]
2:	cnfg(p(bbr2, nb, #b000, ns), pc, semb, sem(sc, dc, tc), sems, fs, #b0) → cnfg(p(bbr2, nb, tc, ns), pc, semb, sem(sc, dc, tc2), sems, fs, #b0)	[sc = #b0 ∧ tc2 = tcbvadd#b010]
3:	cnfg(p(bbr2, nb, nc, ns), pc, semb, sem(sc, dc, tc), sems, fs, #b1) → cnfg(p(bbr3, nb, #b000, ns), pc, semb, sem(sc, dc, tc), sems, fs, #b0)	[¬nc = #b000 ∧ nc = dc]
4:	cnfg(p(bbr3, nb, nc, #b000), pc, semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(p(bbr4, nb, nc, #b000), pc, semb, semc, sem(ss1, ds, ts), fs, #b0)	[¬ss = #b0 ∧ ss1 = ssbvsub#b1]
5:	cnfg(p(bbr3, nb, nc, #b000), pc, semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(p(bbr3, nb, nc, ts), pc, semb, semc, sem(ss, ds, ts2), fs, #b0)	[ss = #b0 ∧ ts2 = tsbvadd#b010]
6:	cnfg(p(bbr3, nb, nc, ns), pc, semb, semc, sem(ss, ds, ts), fs, #b1) → cnfg(p(bbr4, nb, nc, #b000), pc, semb, semc, sem(ss, ds, ts), fs, #b0)	[¬ns = #b000 ∧ ns = ds]
7:	cnfg(p(bbr4, nb, nc, ns), pc, semb, semc, sems, fs, #b0) → cnfg(p(bbr5, nb, nc, ns), pc, semb, semc, sems, fs1, #b0)	[fs1 = (fsbvadd#b1)]
8:	cnfg(p(bbr5, #b000, nc, ns), pc, sem(sb, db, tb), semc, sems, fs, #b0) → cnfg(p(bbr6, #b000, nc, ns), pc, sem(sb, db2, tb), semc, sems, fs, #b1)	[¬tb = dbbvadd#b010 ∧ db2 = dbbvadd#b010]
9:	cnfg(p(bbr5, #b000, nc, ns), pc, sem(sb, db, tb), semc, sems, fs, #b0) → cnfg(p(bbr6, #b000, nc, ns), pc, sem(sb1, db, tb), semc, sems, fs, #b0)	[tb = dbbvadd#b010 ∧ sb1 = sbbvadd#b1]
10:	cnfg(p(bbr6, nb, nc, #b000), pc, semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(p(bbr2, nb, nc, #b000), pc, semb, semc, sem(ss, ds2, ts), fs, #b1)	[¬ts = dsbvadd#b010 ∧ ds2 = dsbvadd#b010]
11:	cnfg(p(bbr6, nb, nc, #b000), pc, semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(p(bbr2, nb, nc, #b000), pc, semb, semc, sem(ss1, ds, ts), fs, #b0)	[ts = dsbvadd#b010 ∧ ss1 = ssbvadd#b1]
12:	cnfg(pb, p(cmr2, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr3, nb, nc, #b000), semb, semc, sem(ss1, ds, ts), fs, #b0)	[¬ss = #b0 ∧ ss1 = ssbvsub#b1]
13:	cnfg(pb, p(cmr2, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr2, nb, nc, ts), semb, semc, sem(ss, ds, ts2), fs, #b0)	[ss = #b0 ∧ ts2 = tsbvadd#b010]
14:	cnfg(pb, p(cmr2, nb, nc, ns), semb, semc, sem(ss, ds, ts), fs, #b1) → cnfg(pb, p(cmr3, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0)	[¬ns = #b000 ∧ ns = ds]
15:	cnfg(pb, p(cmr3, nb, nc, ns), semb, semc, sems, fs, #b0) → cnfg(pb, p(cmr4, nb, nc, ns), semb, semc, sems, fs, #b0)	[fsbvugt#b0]
16:	cnfg(pb, p(cmr3, nb, nc, ns), semb, semc, sems, fs, #b0) → cnfg(pb, p(cmr11, nb, nc, ns), semb, semc, sems, fs, #b0)	[fsbvule#b0]
17:	cnfg(pb, p(cmr4, nb, nc, ns), semb, semc, sems, fs, #b0) → cnfg(pb, p(cmr5, nb, nc, ns), semb, semc, sems, fs1, #b0)	[fs1 = (fsbvsub#b1)]
18:	cnfg(pb, p(cmr5, nb, #b000, ns), semb, sem(sc, dc, tc), sems, fs, #b0) → cnfg(pb, p(cmr6, nb, #b000, ns), semb, sem(sc, dc2, tc), sems, fs, #b1)	[¬tc = dcbvadd#b010 ∧ dc2 = dcbvadd#b010]
19:	cnfg(pb, p(cmr5, nb, #b000, ns), semb, sem(sc, dc, tc), sems, fs, #b0) → cnfg(pb, p(cmr6, nb, #b000, ns), semb, sem(sc1, dc, tc), sems, fs, #b0)	[tc = dcbvadd#b010 ∧ sc1 = scbvadd#b1]
20:	cnfg(pb, p(cmr6, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr7, nb, nc, #b000), semb, semc, sem(ss, ds2, ts), fs, #b1)	[¬ts = dsbvadd#b010 ∧ ds2 = dsbvadd#b010]
21:	cnfg(pb, p(cmr6, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr7, nb, nc, #b000), semb, semc, sem(ss1, ds, ts), fs, #b0)	[ts = dsbvadd#b010 ∧ ss1 = ssbvadd#b1]
22:	cnfg(pb, p(cmr7, #b000, nc, ns), sem(sb, db, tb), semc, sems, fs, #b0) → cnfg(pb, p(cmr2, #b000, nc, ns), sem(sb1, db, tb), semc, sems, fs, #b0)	[¬sb = #b0 ∧ sb1 = ssbvsub#b1]
23:	cnfg(pb, p(cmr7, #b000, nc, ns), sem(sb, db, tb), semc, sems, fs, #b0) → cnfg(pb, p(cmr7, tb, nc, ns), sem(sb, db, tb2), semc, sems, fs, #b0)	[sb = #b0 ∧ tb2 = tbbvadd#b010]
24:	cnfg(pb, p(cmr7, nb, nc, ns), sem(sb, db, tb), semc, sems, fs, #b1) → cnfg(pb, p(cmr2, #b000, nc, ns), sem(sb, db, tb), semc, sems, fs, #b0)	[¬nb = #b000 ∧ nb = db]
25:	cnfg(pb, p(cmr11, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr2, nb, nc, #b000), semb, semc, sem(ss, ds2, ts), fs, #b1)	[¬ts = dsbvadd#b010 ∧ ds2 = dsbvadd#b010]
26:	cnfg(pb, p(cmr11, nb, nc, #b000), semb, semc, sem(ss, ds, ts), fs, #b0) → cnfg(pb, p(cmr2, nb, nc, #b000), semb, semc, sem(ss1, ds, ts), fs, #b0)	[ts = dsbvadd#b010 ∧ ss1 = ssbvadd#b1]
27:	cnfg(pb, pc, semb, semc, sems, fs, a) → success	
28:	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
29:	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
30:	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr3, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
31:	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
32:	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
33:	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr4, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
34:	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
35:	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
36:	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr5, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
37:	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
38:	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
39:	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr6, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
40:	cnfg(p(bbr4, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
41:	cnfg(p(bbr5, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	
42:	cnfg(p(bbr6, nbb, nbc, nbs), p(cmr11, ncb, ncc, ncs), semb, semc, sems, fs, a) → error	

$$\{abbcbp(bb2\#b000\#b000\#b000), sem(\#b0, d, b), sem(\#b0, d, t), sem(\#b1, d, s), \#b1, \#b0) \Rightarrow succs [tb = db \vee add\#b010 \wedge (db \vee gt\#b000) \wedge ts = ds \vee add\#b010 \wedge (ds \vee gt\#b000)]$$